Auction Bidding System

A GUI-BASED PYTHON PROJECT WITH POSTGRESQL INTEGRATION

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◇ 1. Introduction

The Auction Bidding System is a Python-based application that enables users to participate in online bidding through a graphical user interface. The system maintains real-time data, bidding history, and winner declaration using a PostgreSQL database.

◇ 2. Objective

To simulate a digital auction house allowing multiple users to view, bid on items, and automatically determine the winner based on the highest bid before the timer ends.

Module Name	Description
User Login & Registration	Secure authentication for users.
Dashboard	User panel with action buttons and views.
Auction Timer	Tracks remaining time for each auction.
Winner Declaration	Automatically picks the highest bidder.
Bidding System	Accepts and validates user bids.
Database Integration	PostgreSQL-backed real-time updates.

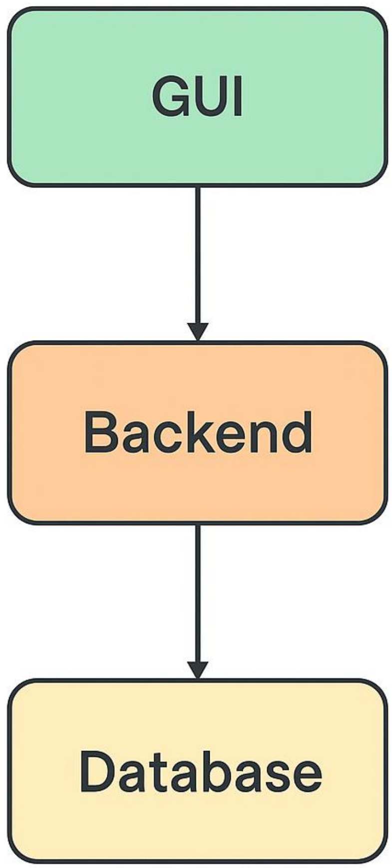
◇ 4. Team Members & Contributions

Team Member	Contribution Area
Shravika	Login and Registration
Harman	Timer & Winner Declaration
Sarthak	Dashboard & Database Integration
Aaradhya	Bidding Functionality

◇ 5. Technical Stack

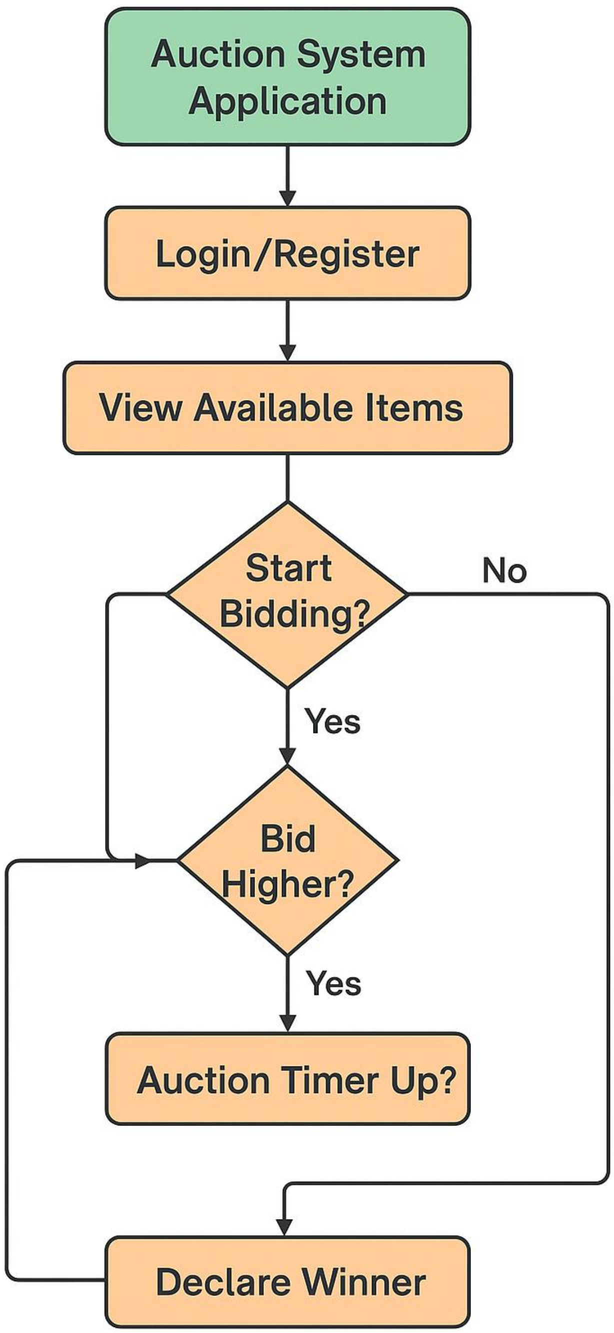
- **Language**: Python
- **GUI**: Tkinter
- **Database**: PostgreSQL
- **Libraries**: tkinter, datetime, psycopg2

◇ **6. System Architecture**



System Architecture

- **7. Bidding Flowchart**



8. Key Functionalities

- ☑ Real-time auction timer
- ☑ Secure bid placement with validation
- ☑ Winner popup notification
- ☑ User-specific and all bids view

9. Future Improvements

- Admin panel for item upload
- SMS/Email notifications to winners
- Integration with payment gateway
- Hashing of passwords for enhanced security

10. Conclusion

This project demonstrated a practical application of GUI programming, database integration, and team collaboration. The modular development allowed smooth progress and efficient testing of each component.